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1 October 1990
SUPERSEDING
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2 December 1974

MILITARY SPECIFICATION

GRINDING MACHINES, SURFACE, ROTARY TABLE, HORIZONTAL SPINDLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, rotary table, horizontal spindle, plain and automatic, surface grinding machines with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Government's minimum requirement for each procurement.

1.2 Classification. The grinding machines shall be of the following types and of the sizes listed in table I. The types and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Vertical wheelhead column

Type II - Horizontal wheelhead ram

Type III - Crossrail mounted wheelhead

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook. Tolerances and Measuring Methods for Unassembled Spur and Helical Gears. (Including metric equivalents) (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DOD adopted)

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B7.1 - Safety requirements for the Use, Care, and Protection of Abrasive Wheels. (DOD adopted)

ANSI B11.9 - Safety Requirements for the Construction, Care, and Use of Grinding Machines. (DOD adopted)

ANSI B74.13 - Markings for Identifying Grinding Wheels and Other Bonded Abrasives. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

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(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete grinding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The grinding machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The design of the type I machines shall feature an adjustable wheelhead assembly mounted on a vertical column with horizontal reciprocation of a single rotating chuck or worktable. Type II machines shall incorporate in their design a horizontal wheelhead and ram assembly with vertical adjustment of the chuck or worktable. The design of the type III machines shall include a column with vertical ways, on which is mounted a crossrail, thereby providing vertical travel. A horizontal wheelhead shall be mounted on guideways provided by the crossrail for traverse feed. The design and construction of the machine shall be in accordance with either the U.S. Customary System of Units (US) or the International System of Units (SI).

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

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3.2.1.1 Dials. All dials used to indicate machine linear axes position or tool movement shall be graduated in increments not larger than 0.0001 inch for the inch system, not larger than 0.003mm for the metric system, or not larger than 0.0002 inch and 0.005mm for the dual system of measurements. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the Y and Z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0005 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be half of the resolution, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input source fluctuation of $\pm$ 10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable from a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so

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as to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal machine operation.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The grinding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machines.

3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. An automatic lubrication system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

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3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts, such as ways, shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard shall also be considered replacement parts. Replacement parts shall be domestically available.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. Any excess material used in such processes shall be thoroughly removed from the part(s) upon completion of the operation.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2000-A88 and AGMA 2001-B88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. Drive train gears shall be manufactured of a suitable ferrous material and of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

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3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The grinding machine shall consist basically of the following components. Any additional components necessary for the operation of the machine shall be furnished in accordance with the manufacturer's standard practice.

3.4.1 Bed. The bed shall be of a material sufficient in strength and rigidity to withstand all forces encountered during operation of the machine to its maximum capacity while maintaining mutual component alignment as necessary for meeting performance, accuracy, and alignment requirements stated herein. The type I machine bed shall have horizontal guideways to accept a reciprocating worktable or carriage. The guideways shall be of sufficient length and area to support the loaded table or carriage and magnetic chuck and convey them from the loading position to the grinding position. The type II bed shall have horizontal guideways to accept a ram and wheelhead assembly. The type III bed shall have precision circular guideways to accommodate a non-traversing, rotating worktable and provide for mounting a vertical column and its components. When hydraulic or coolant reservoirs are housed in the bed, access door(s) or removable plate(s) shall be provided for servicing components. Leveling screws and floor mounts shall be provided around the periphery of the base.

3.4.2 Vertical column (type I machines). The vertical column on the type I machines shall be ribbed and braced to minimize distortion and deflection under normal workloads. The column shall be arranged to accommodate a vertically traversing grinding wheelhead assembly and its components. Ground or hand scraped guideways shall be provided to allow vertical travel of the grinding wheel assembly. Provisions shall be made to compensate for guideway wear.

3.4.3 Horizontal ram (type II machines). Each type II machine shall be provided with a ram and wheelhead assembly arranged to reciprocate horizontally on guideways of the bed. The assembly slides shall accurately fit mating surfaces of the bed.

3.4.4 Vertical column with crossrail (type III machines). The vertical column on the type III machines shall be ribbed and braced to minimize distortion and deflection under normal workloads. The column shall have precision guideways to accommodate a vertically traversing crossrail and its components.

3.4.4.1 Crossrail. The tilting crossrail on the type III machines shall have precision horizontal guideways to accommodate the traversing wheelhead assembly. The crossrail shall be complete with a graduated indicator and locking device, to accomplish convex and concave work. The amount of tilt shall be in accordance with table I.

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3.4.5 Worktable. The worktable on each machine shall be proportioned to withstand any loads that may be encountered during normal operations. The worktable shall be constructed to accept a magnetic chuck arranged to have rotating motion on a vertical axis. A rotating mechanism, complete with feed and speed components, drive mechanism, and controls, as applicable, shall be provided. Each worktable on the type I and the type II machines shall be provided with a swivel arrangement, complete with graduated indicator and locking device, to accomplish concave and convex work. The worktable on type I machines shall be mounted on guideways of the bed and shall traverse in a longitudinal direction; the worktable on the type II machines shall traverse vertically on an arrangement provided by the bed. The worktable on the type III machines shall rotate on a vertical axis with no provisions for vertical or longitudinal movement. Unless otherwise specified (see 6.2), power traverse shall be provided with means for manual control. Provisions shall be made for locking the table or carriage after positioning.

3.4.6 Magnetic chuck. An electro-magnetic chuck, not less than the diameter of the table or carriage, shall be provided with all accessories necessary for installation and operation, including necessary equipment to magnetize and demagnetize the chuck. The chuck shall have full and variable holding power. The chuck shall be sealed to prevent fluid leakage into the coils or accumulation of moisture therein. The poles of the chuck shall be not more than 1/4 inch apart over the entire face of the chuck to ensure small parts as well as large parts will be held securely during grinding operations. The chuck shall operate from the same power source as the machine and shall be completely wired and connected to the main power panel on the machine. When specified (see 6.2), a safety interlock shall be provided to prevent spindle rotation unless the chuck is energized. Different chuck diameter and other special chuck features shall be as specified (see 6.2).

3.4.7 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a wheelhead housing, spindle and spindle driving mechanism. The spindle driving mechanism shall be designed to operate at the RPM required in table I. The spindle shall be dynamically balanced and mounted in precision anti-friction bearings designed to minimize axial and radial play. The spindle shall be constructed to accept a driving mechanism at one end and a grinding wheel and holder at the other end. The driving mechanism shall be either an electric motor with belt and pulley arrangement, direct electric motor drive, or hydraulic drive. Wheelhead and hydraulic motors may be combined provided the total horsepower meets the requirements of table I. The spindle shall be mounted in precision anti-friction bearings provided with continuous or permanent lubrication. The assembly shall be mounted on guideways provided by the column, ram or crossrail and shall traverse vertically or horizontally as applicable. The assembly shall have both automatic and manual movement for downfeed and infeed of the grinding wheel. Downfeed and infeed shall be controlled by pre-determined stops.

3.4.7.1 Manual feed. A handwheel or other manually operated device shall be furnished for accomplishing workpiece surface to grinding wheel contact. The system shall have a coarse and fine feed with an indicator graduated in accordance with 3.2.1.1.

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3.4.7.2 Power feed. Each machine shall be designed with power feed of the abrasive wheel or workpiece. Selectors shall be provided for controlling and limiting the total feed for the overall grinding cycle.

3.4.8 Semi-automatic grinding cycle. When specified (see 6.2), a semi-automatic grinding system shall be provided and fully described. After set-up, the operator shall only be required to load the workpiece, close the circuit to the chuck, start the cycle by a single control and unload the workpiece. The grinding cycle shall include but not be limited to the following functions:

A - Rapid advance, chuck starts rotating
B - Wheel feeds at outside of work as required
C - Grind in as required
D - Wheel feeds at inside of work as required
E - Grind out as required
F - Table or wheel returns at grind speed
G - Rapid return, chuck stops

3.4.9 Automatic grinding system. When specified (see 6.2), except for sizes 6, 8 and 30, an automatic grinding cycle shall be provided. After the machine is started, the only action required of the operator shall be to keep the hopper or other parts feed device supplied with parts. The machine shall automatically receive and position the part, grind to size and eject the part. The system shall provide for automatic wheel dressing and compensation for wheel size reduction, automatic feed, and automatic spark out.

3.4.10 Wheel dresser. Unless otherwise specified (see 6.2), a permanently mounted, manually operated, wheel dresser shall be provided, capable of dressing the entire periphery of the grinding wheel.

3.4.11 Grinding wheels. Required grinding wheels shall be as specified (see 6.2). Grinding wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All wheels shall be properly guarded.

3.4.12 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump, and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant over the entire work area. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow.

3.4.13 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level and provisions for draining.

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3.4.14 Electrical system. The electrical system shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed for operation on 3-phase, 60-hertz (Hz), dual 230/460 voltage. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.14.1 Motors. Motors shall meet the requirements of 3.4.13. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.14.2 Worklight. Unless otherwise specified (see 6.2), a worklight conforming to 3.4.13 shall be provided. The light shall have a protective shield and an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.14.3 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the grinding machine shall meet the requirements of table I for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.6 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

3.7 Performance. The machine shall be capable of successfully performing all tests specified in 4.8.2 through 4.8.4. Performance test results shall be within those in 4.8.4.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

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3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the grinding machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

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4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The grinding machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The grinding machine shall be operated at no load for not less than 2 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance tests given in 4.8.4 through 4.8.4.1. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance tests. The machine shall be tested to determine its capability to rough and finish grind both top and bottom surfaces of eight blocks, four placed within 1 inch of the outer periphery of the chuck, spaced equally apart and four placed within 1 inch of the chuck center or pilot hole, spaced equally apart. The test blocks shall be of SAE 1040 steel, size and thickness at the supplier's option. The test blocks shall be finish ground to 10 microinches or better. All finish test blocks shall be of uniform thickness within 0.0001 inch, or 0.0001 inch per foot of their separation on the chuck, whichever is larger. The methods used to obtain the above designated requirements shall be at the supplier's option. A ring test part may be used in lieu of test blocks. If used, the test rings shall meet the above requirements.

4.8.4.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

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5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in production shops where rapid metal removal during batch grinding of like workparts is desired or in toolroom applications where precision work with a superior finish is required, including concave and convex grinding.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Digital readout system, it not required (see 3.2.1.2).

g. Specify any special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Power traverse, if different (see 3.4.5).

m. Safety interlock preventing spindle rotation, if required (see 3.4.6).

n. Fully describe different chuck size and other special chuck features, if required (see 3.4.6).

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o. Semi-automatic grinding system, specify and fully describe (see 3.4.8).

p. Automatic grinding system, if required (see 3.4.9).

q. Wheel dresser, if different (see 3.4.10).

r. Describe required grinding wheels (see 3.4.11).

s. Electrical system, if different (see 3.4.14).

t. Specify initial voltage, (see 3.4.14).

u. Motors, if different (see 3.4.14.1).

v. Worklight, if not required (see 3.4.14.2).

w. Hour meter, if required (see 3.4.14.3).

x. Size and capacity, if different (see 3.5).

y. Additional performance features, if required (see 3.7.1).

z. Specify and fully describe optional equipment, if required (see 3.9).

aa. Specify applicable repair parts, if required (see 3.10).

bb. Lubrication plate, if required (see 3.11.1).

cc. Information to be provided on lubrication plate, if different (see 3.11.1).

dd. Nameplate, if different (see 3.11.2).

ee. Hydraulic plate, if required (see 3.11.3).

ff. Hydraulic plate, if different (see 3.11.3).

gg. Inspection conditions, if different (see 4.3).

hh. First article inspection, if different (see 4.4).

ii. Quality conformance inspection, if different (see 4.5).

jj. Acceptance test, if different (see 4.6).

kk. Additional performance tests, if required (see 4.8.4.1).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.13	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language
3.13	DI-TMSS-80528	Supplemental Data For Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-G-80163A		MIL-G-80163B		
Type I,	Size 6	Type I,	Size 6	
	Size 8		Size 8	
	Size 12		Size 12	
	Size 16		Size 16	
	Size 24		Size 24	
	Size 30		Size 30	
Type II,	Size 16	Type II,	Size 16	
	Size 24		Size 24	
	Size 30		Size 30	
	Size 40		Size 40	
		Type III,	Size 20	New
			Size 24	New
			Size 30	New
			Size 40	New

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Hazardous materials minimization. The manufacturer should minimize the use, generation, or emission of materials known hazardous to the environment during the manufacturing process employed at his facilities. During the production of the end item, the manufacturer should comply with the applicable Federal, state, and local statutes in effect at the point of production which regulate the use of hazardous materials. Where practical, alternative materials or processes should be employed in lieu of known hazardous materials or processes to minimize the threat to the environment.

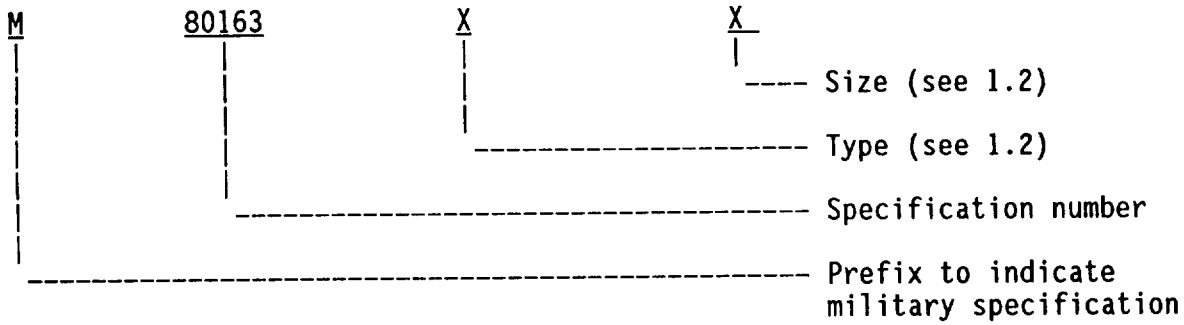
6.11 Training. Training required by the procuring activity should be provided as specified in the contract.

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6.12 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.13 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.14 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification is created as follows:



6.15 Subject term (keyword) listing.

Concave
Convex
Industrial plant equipment
Machine shop
Machine tool
Metal working
Precision finish
Precision tolerance

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TABLE I. Sizes and capacities.

Characteristic	Requirements					
	Type I					
	Size 6	Size 8	Size 12	Size 16	Size 24	Size 30
Capacities						
Chuck or table diameter, inches	6	8	12	16	24	30
Swing (inside of coolant pan) inches	12	12-1/2	18	18	28	32
Standard wheel size, inches	<----7x1/2x1-1/4--->		14x1x5	14x1x5	14x1-1/2x5	18x1-1/2x8
Work height , inches, new wheel	6-1/2	6-1/2	10	10	10	8-1/2
worn wheel	7-1/4	7-1/4	11-1/2	11-1/2	12	9
Angular tilt, degrees, concave	15	15	10	10	10	10
convex	15	15	17	17	17	17
Chuck or table travel, inches,						
horizontal	6	6-1/4	12	12	16	16
vertical	--	--	--	--	--	--
*Grinding stroke, inches	4	4	8	8	12	15
Speeds and feeds						
Chuck or table speeds, number	1	1	<----------infinite---------->			
Chuck or table speed, or range, RPM	350	350	50 to 225	60 to 225	38 to 100	30 to 80
Wheel speed, RPM	3600	3600	1640	1640	1765	1300
Wheelhead traverse rate range,						
vertical, IPM	--	--	0 to 6	0 to 6	0 to 6	0 to 6
horizontal, FPM	--	--	--	--	--	--
Chuck or table traverse speed range,						
horizontal, FPM	0 to 10	0 to 10	0 to 12	0 to 12	0 to 12	0 to 15
vertical, IPM	--	--	--	--	--	--
Power feed increment range,	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
inches per increment	to	to		to	to	to
	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
Motors						
Wheelhead drive, H.P.	1-1/2	1-1/2	--	--	15	15
Hydraulic drive, H.P.	1/3	1/3	3	3	5	5
Coolant pump, H.P.	1/10	1/10	1/2	1/2	1/2	1/2
Chuck or table drive, H.P.	1/4	1/4	--	--	--	--

*Unless otherwise specified (see 6.2).

Note 1: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in the above table. When the largest size and capacity machine shown in the above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent.

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TABLE I. Sizes and capacities - continued.

Characteristic	Requirements							
	Type II				Type III			
	Size 16	Size 24	Size 30	Size 40	Size 19	Size 23	Size 29	Size 39
Capacities								
Chuck or table diameter, inches	16	24	30	40	19	23	29	39
Swing (inside of coolant pan) inches	26	34	34	47	29	30	32	46
Standard wheel size, inches	16x2x8	16x2x8	16x2x8	16x2x8	15x2x5	15x2x5	15x2x5	17x3-1/2x7
Work height , inches, new wheel	10	10	10	10	25	25	25	27
worn wheel	12	12	12	12	27	27	27	32
Angular tilt, degrees, concave	10	10	10	3	15	15	15	15
convex	10	10	10	3	15	15	15	15
Wheelhead travel, inches, vertical	--	--	--	--	27	27	27	32
Chuck or table travel, inches,								
vertical	12	12	12	12	--	--	--	--
*Grinding stroke, inches	13	17	17	22	11-1/2	13-1/2	15-1/2	21-1/4
Speeds and feeds								
Chuck or table speeds, number	<----------infinite---------->				<----------infinite---------->			
Chuck or table speed, or range, RPM	60 to 250	37 to 155	30 to 125	25 to 100	60 to 250	37 to 155	30 to 125	25 to 100
Wheel speed, RPM	1400	1400	1400	1400	1400	1400	1400	1400
Wheelhead traverse rate range,								
vertical, IPM	--	--	--	--	0 to 10	0 to 10	0 to 10	0 to 10
horizontal, FPM	0 to 10	0 to 10	0 to 10	0 to 10	0 to 10	0 to 10	0 to 10	0 to 10
Worktable traverse speed range,								
horizontal, FPM	--	--	--	--	--	--	--	--
vertical, IPM	0 to 6	0 to 6	0 to 6	0 to 6	--	--	--	--
Power feed increment range,	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
inches per increment	to	to	to	to	to	to	to	to
	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015	0.0015
Motors								
Wheelhead drive, H.P.	20	20	20	20	15	15	15	20
Hydraulic drive, H.P.	7-1/2	10	10	10	7-1/2	7-1/2	7-1/2	10
Coolant pump, H.P.	1/4	1/4	1/4	1/4	1/4	1/4	1/4	1/4
Chuck or table drive, H.P.	5	5	5	5	5	5	5	5

*Unless otherwise specified (see 6.2).

Note 1: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in the above table. When the largest size and capacity machine shown in the above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent.

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TABLE II. Alignment tolerances

Test to be applied	Maximum Allowable Deviation
Worktable flat in all directions, through center	0.0001/12"
Axial runout of worktable	0.0001" TIR
Surface of worktable parallel with its transverse movement	0.0001/12"
Wheelhead spindle radial runout	0.0002" TIR
Wheelhead spindle axial slip	0.0002" TIR
Vertical adjustment of wheelhead assembly square with worktable, Y and Z axes	
Sizes 6 and 8 machines	0.004/12"
All other sizes	0.006/12"

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Navy - MC

Preparing activity:
DLA - IP

Project (3415-0183)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-80163B	901001

3. DOCUMENT TITLE
Grinding Machines, Surface, Rotary Table, Horizontal Spindle

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM, Memphis, TN 38114	(1) Commercial (901) 775-4797	(2) AUTOVON 683-4797
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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